Multi-singlet dynamics in complex double-scaled SYK
A reduction to mixed chord moments

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Abstract

After subtracting its conserved quadratic component, the first derivative singlet has the joint moments of a normalized chord probe of length ratio two. We extend the two-insertion endpoint argument to any fixed number of insertions. The resulting crossing matrix has entries q, q^2, q^4. An explicit connected four-point function follows from the known q-Mehler kernel; its equal-time value is $e^{-16\lambda} - 1$. The reduction supplies a quantitative interaction diagnostic for the singlet sector of the proposed static-patch hologram.

1 The singlet and the joint-moment result

Use standard CAR, $\hbar = 1$, string-time Hamiltonian $H = H_s$, and state $\tau_N = \mathbb{E}\, 2^{-N}\,\mathrm{Tr}$ at $\beta = \mu = 0$:

$$H = \sum_{|I|=|J|=p} G_{IJ} c_I^\dagger c_J, \qquad \mathbb{E} G_{IJ} G_{KL} = w \delta_{IL} \delta_{JK}, \tag{1}$$

$$w = 4^p \frac{\mathcal{J}_s^2}{\lambda} \left(\frac{p!}{N^p} \right)^2, \qquad \lambda = p^2/N, \quad q = e^{-4\lambda}, \quad \sigma^2 = \mathcal{J}_s^2/\lambda. \tag{2}$$

Overlapping index sets are included in the full Hermitian Gaussian ensemble. All limits first take $N, p \to \infty$ at fixed $\lambda > 0$. Define

$$A = \sum_i [H, c_i]^\dagger [H, c_i], \qquad O = A - \mathbb{E} A(Q), \qquad B = H^2 - \mathbb{E} H^2(Q), \tag{3}$$

$$c = 2\lambda \frac{1-q}{1+q}, \qquad W_d = 4\mathcal{J}_s^4 \left(\lambda^{-1} + \frac{4q}{1+q} \right), \qquad X_N = \frac{O - cB}{\sqrt{W_d}}. \tag{4}$$

Here Q is fermion number and B is conserved. One may also use the finite-system projection coefficient $c_N = \tau_N(OB)/\tau_N(B^2)$: $c_N \to c$, and all results below are unchanged.

Joint-moment reduction. For every finite word in H/σ and X_N, its limiting moment is the sum over pairings of equal letters, with a factor for each crossing:

$$\boxed{\mathcal{Q}_{HH} = q, \qquad \mathcal{Q}_{HX} = q^2, \qquad \mathcal{Q}_{XX} = q^4.} \tag{5}$$

These are established mixed q-Gaussian/chord moments [3, 1, 2]. Our calculation identifies this same-coupling composite with that known moment system; it does not replace the underlying Gaussian couplings by independent ones at finite N.

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2 An explicit connected four-point function

Write τ for the limiting tracial state and $X(t) = e^{itH} X e^{-itH}$. All correlators with an odd number of X insertions vanish, including the three-point function at arbitrary fixed real times. At equal time,

$$\tau(X^2) = 1, \qquad \tau(X^4) = 2 + q^4, \qquad \tau(X^6) = 5 + 6q^4 + 3q^8 + q^{12}. \tag{6}$$

The three pairings of four X endpoints have weights $1, 1, q^4$. Thus the ordinary fourth cumulant is $q^4 - 1$, rather than zero.

Let $G_r(t)$ denote the normalized known probe kernel [1, 2]:

$$E(\theta) = \frac{2\sigma\cos\theta}{\sqrt{1-q}}, \qquad \rho_q(\theta) = \frac{(q, e^{2i\theta}, e^{-2i\theta}; q)_\infty}{2\pi}, \tag{7}$$

$$K_r(\theta, \phi) = \frac{(r^2; q)_\infty}{\prod_{\epsilon,\eta=\pm1}(re^{i(\epsilon\theta+\eta\phi)}; q)_\infty}, \tag{8}$$

$$G_r(t) = \int_0^\pi \int_0^\pi \mathrm{d}\theta\, \mathrm{d}\phi\, \rho_q(\theta)\rho_q(\phi) K_r(\theta, \phi) e^{it[E(\theta)-E(\phi)]}. \tag{9}$$

The notation $(a_1, \ldots, a_k; q)_\infty$ denotes a product of q-Pochhammer symbols. We have $G_r(0) = 1$ and $\tau(X(t)X) = G_{q^2}(t)$.

Consider the consecutive-time arrangement $F_{22}(t) = \tau(X(t)^2 X(0)^2)$. Internal X pairing gives 1. The two remaining pairings connect the two adjacent blocks, with combined weight $1 + q^4$. Every H chord crossing that bundle crosses two X chords and receives q^4. Hence

$$\boxed{F_{22}(t) = 1 + (1 + q^4) G_{q^4}(t).} \tag{10}$$

Subtracting all three products of two-point functions gives

$$\boxed{\kappa_4(t) = F_{22}(t) - 1 - 2G_{q^2}(t)^2 = (1 + q^4) G_{q^4}(t) - 2G_{q^2}(t)^2.} \tag{11}$$

This is an ordinary connected correlator of the annealed tracial state. It is not the out-of-time-order correlator $\tau(X(t)XX(t)X)$.

Two analytic interaction diagnostics. Using $G_r''(0) = -2\sigma^2(1 - r)$ gives

$$\kappa_4(0) = e^{-16\lambda} - 1 = -16\lambda + O(\lambda^2), \tag{12}$$

$$\kappa_4''(0) = 2\sigma^2(3 - 4q^2 + q^8) = 768\mathcal{J}_s^2\lambda + O(\mathcal{J}_s^2\lambda^2). \tag{13}$$

Their leading scale is compatible with $g_{\mathrm{open}}^2 = \lambda$ [7]. They measure departure from ordinary Gaussian pairing; extracting an on-shell bulk coupling still requires the bulk dictionary.

3 Time dependence and the general four-point interface

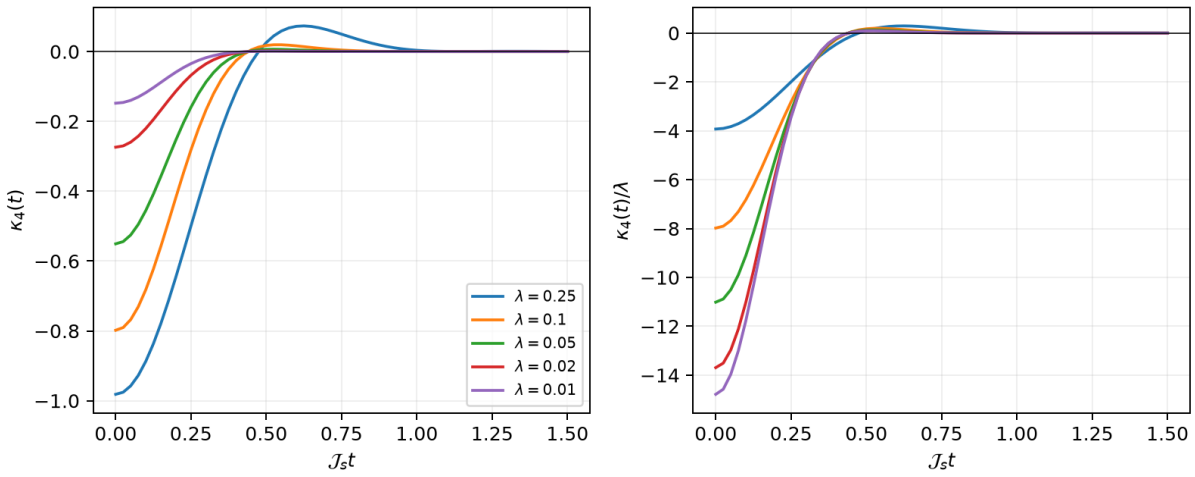


Figure 1. Equation (11) in string units. The right panel rescales by λ; it is not a fitted small-λ formula. Each curve is negative at $t = 0$ and positive at $\mathcal{J}_s t = 0.5$. Continuity therefore ensures at least one zero in between.

λ	$\kappa_4(0)$	$\kappa_4(0.5/\mathcal{J}_s)$	Jacobi dimension
0.25	−0.981684	0.0258033	42
0.10	−0.798103	0.0175599	69
0.05	−0.550671	0.00638964	111
0.02	−0.273851	0.00194250	235
0.01	−0.147856	0.000880010	439

The known fixed-time small-deformation probe limit [4], with $-\log q = 4\lambda$ and energy scale $2\mathcal{J}_s$, gives $G_{q^d}(t) \to \mathrm{sech}^{2d}(2\mathcal{J}_s t)$. Consequently $\kappa_4(t) \to 0$. The first correction to the two kernels must also be retained to find the coefficient of λ at general time; expanding only $1 + q^4$ does not do so.

Four separate times. The entire four-point function is determined by (5), not just the coincident pairs in (10). In the established BINT formulas [1], set both Hamiltonian–probe crossing parameters to q^2 and restore the mutual-probe factor $\widetilde{q}_{12} = q^4$. Their Section 4.2 explicitly suppresses this overall factor for a selected crossed channel. It must be restored before adding that channel to the two uncrossed channels for a single species. BNR Section 8.1 [2] gives the corresponding complex-model dictionary.

For clarity an alternative exact specification uses chord moments P. With $t_5 = t_1$ and $s_j = t_{j+1} - t_j$,

$$\tau\left(\prod_{j=1}^{4} X(t_j)\right) = \sum_{n_1,\ldots,n_4 \geq 0} \prod_{j=1}^{4} \frac{(i\sigma s_j)^{n_j}}{n_j!}\, P(XH^{n_1}XH^{n_2}XH^{n_3}XH^{n_4}). \tag{14}$$

Here H inside P is unit variance. This series is absolutely convergent in the bounded limiting model. We evaluate the collapsed slice in this note; numerical evaluation of the general crossed kernel is the next task.

A Marked local determinants and the fixed-order limit

Let a word have m composite insertions and n ordinary H slots. Expand each composite into two adjacent slots, giving $L = n + 2m$. The algebraic identity

$$A = (2p - \mathcal{D})H^2, \qquad \mathcal{D}(Z) = QZ - \sum_i c_i^\dagger Z c_i \tag{15}$$

expresses A as a normal-order contraction marker in the adjacent pair. Write its pairs as (a_j, b_j), $b_j = a_j + 1$, with variables h_j. For a Gaussian pairing π of the L Hamiltonian slots use

$$F_\pi(z,h) = \det(I + ZT(h)P_\pi), \quad T_{ab}(0) = \begin{cases} 1/2 & a \le b, \\ -1/2 & a > b, \end{cases} \quad T_{b_j a_j}(h) = T_{b_j a_j}(0) - h_j. \tag{16}$$

With $Z = \mathrm{diag}(z_1, \dots, z_L)$ the finite coefficient is

$$w^{L/2}(-1)^{Lp/2}[z_1^p \cdots z_L^p] \prod_{j=1}^{m} (\partial_{h_j} - c) F_\pi(z,h)^N \Big|_{h=0}. \tag{17}$$

Centering each quadratic Gaussian block deletes pairings internal to that block. This is exact before taking any limit, including for the $-cB$ terms. Therefore no surviving singleton minor contains a marker.

For fixed L, pair overlaps of the uniform p-sets have the joint Poisson limit with mean λ. Triple membership is bounded by $\binom{L}{3}p^3/N^2 = o(1)$. The hypergeometric estimate

$$\mathbb{E}e^{s|I\cap J|} \le \exp\left[\frac{p^2}{N-p+1}(e^s - 1)\right], \qquad s \ge 0, \tag{18}$$

together with Hölder bounds controls all finitely many overlaps. After division by singleton factors, the local marked minors are bounded on a fixed complex neighborhood. Their products are dominated by an exponential in total pair overlap. Uniform integrability removes the triple events; Cauchy's formula passes any fixed set of mark derivatives through the limit. This is the same fixed-order argument as in the preceding two-point note, now with finitely many marker variables.

Only two-site minors survive, so the marked exponent has degree at most two in the nilpotent variables:

$$\sigma^{n+2m} q^{\mathrm{cr}(\pi)} \exp\left(\lambda \sum_j \alpha_j h_j + \lambda \sum_{i<j} \delta_{ij} h_i h_j\right), \qquad h_j^2 = 0. \tag{19}$$

The local determinant gives $\alpha_j = 2\epsilon_j$, where ϵ_j is $+1$ for noncrossing incident chords and -1 for crossing ones. Also $\delta_{ij} = 4$ precisely when the two marks are fully joined by two nested chords, and is zero otherwise. No uniformity as $m + n \to \infty$ is assumed at this step.

B Why only paired singlet bundles survive

Put $u = (1-q)/(1+q)$. Applying the shifted derivatives in (19) is the finite partial-matching sum with singleton means

$$\ell_j = \lambda(\alpha_j - 2u) \tag{20}$$

and paired factors $\lambda\delta_{ij}$. Suppose a marked block is not connected by both of its incident chords to a single other marked block. Then every δ involving it is zero. Its factor in every term is consequently ℓ_j.

Exchange its two adjacent endpoints. This changes only the crossing of the two incident chords and flips ϵ_j; it changes neither other α's nor any δ's. If another mark shares one of these chords, its other incident chord is distinct, so its crossing sign is unchanged. The paired diagram weights are w and qw, and

$$(2\lambda - c) + q(-2\lambda - c) = 0. \tag{21}$$

The orbit thus cancels exactly. This also covers a triangle or a longer cycle of marked blocks; sharing one chord is insufficient to escape the cancellation.

All surviving marked blocks must therefore be joined into disjoint double-chord pairs. Odd m gives zero. For one pair, nested and crossed configurations sum to

$$D = (2\lambda - c)^2 + 4\lambda + q(-2\lambda - c)^2 \tag{22}$$

$$= 4\lambda + \frac{16\lambda^2 q}{1+q} = W_d/\sigma^4. \tag{23}$$

This factor includes both the local quadratic marker and the correlated singleton means. Keeping only the δ term would give a wrong norm.

Compress each double chord into one X chord. A crossing with one ordinary Hamiltonian chord has two constituent crossings and weight q^2; a crossing between two such bundles has four and weight q^4. The internal choices summed in (23) do not change these external crossings. Dividing by $W_d^{m/2}$ leaves σ^n times the mixed pairing polynomial (5). This proves the joint-moment reduction at every fixed order.

The finite choice c_N gives the same limits: expand each insertion in $c_N - c$ and use $c_N \to c$ and bounded fixed higher moments of O, B, H. The deterministic centering $\mathbb{E}H^2(Q)$ has all fixed moments tending to those of $\sigma^2 I$ (prescribed internal adjacent pairs). Thus there is no hidden random conserved offset in this replacement.

Independent checks. The code computes multi-mark two-site minors using integer nilpotent polynomials, then compares their sums with independent compressed-probe pairings. All 95 words pass, including all placements of three X with four H, four X with two H, and X^6: 41,210 local diagrams in total. This finite check supports, but does not replace, the orbit proof.

C Real-time passage and numerical control

The known mixed q-Gaussian construction [3] realizes (5) in a positive tracial state. At fixed $0 < q < 1$ its single-generator spectral supports give

$$\|H\| \le R_H = \frac{2\sigma}{\sqrt{1-q}}, \qquad \|X\| \le R_X = \frac{2}{\sqrt{1-q^4}}. \tag{24}$$

This makes (14) well defined. To connect it to finite N, let $P_K(tH) = \sum_{k=0}^{K} (itH)^k/k!$. For even s the real-axis Taylor remainder gives, in the annealed noncommutative L^s norm,

$$\limsup_{\mathrm{DS}} \|e^{itH_N} - P_K(tH_N)\|_s \le \frac{|t|^{K+1}}{(K+1)!} \lim_{\mathrm{DS}} \tau_N(|H_N|^{s(K+1)})^{1/s} \le \frac{(|t|R_H)^{K+1}}{(K+1)!}. \tag{25}$$

All fixed even L^s norms of X_N remain bounded and tend to those of X; the polynomial approximants have limiting norms at most $e^{|t|R_H}$. Expand the difference between a correlator and its polynomial approximation by telescoping, and apply tracial Hölder with an even exponent at least the number of factors. Equation (25) sends every error to zero after the double-scaling limit. Polynomial convergence then proves convergence at any fixed finite set of real times, including (10). This does not assume finite-N operator-norm convergence or interchange an infinite moment series with that limit.

Numerical evaluation. The Jacobi off-diagonal entries are $\sigma\sqrt{(1-q^{k+1})/(1-q)}$. For $\psi(t) = e^{-itT}|0\rangle$,

$$G_r(t) = \sum_{k\ge 0} r^k |\psi_k(t)|^2. \tag{26}$$

At dimension L, the first L vacuum-vector Taylor coefficients agree. With $a = |t|R_H$ and $L+1 > a$, the error of either bounded-vertex expectation is at most

$$b_L = \frac{4a^L}{L!} \frac{1}{1 - a/(L+1)}. \tag{27}$$

We choose $b_L < 10^{-13}$ up to $\mathcal{J}_s t = 1.5$. Since both exact and truncated G lie in $[0,1]$, the connected-function truncation error is at most $6b_L$. Dimensions $L+16$ differ by at most 5.2×10^{-15}. Independent 96/192-node Mehler quadratures check both kernels, the second frequency moments and the connected difference at $\lambda = 0.25, 0.1$. The saved records include relative residuals and cancellation condition numbers. These quadrature and floating-point checks are empirical; b_L bounds only the Jacobi truncation. No late-time extrapolation is used.

D Validation, provenance and the next physical calculation

Independent CAR matrices check the finite marked-determinant formula before its Poisson limit: 160 words at $(N, p) = (3, 1), (4, 2)$ with three, four or six marks, and 3,584 exact rational comparisons of all marker subsets. Forty-two fully marked traces are nonzero; none of those are negative. The set includes coherent repeated-index words to avoid testing mostly zeros. Fraction-free Bareiss determinants were checked against Laplace expansion on 140 small integer matrices. The initial factorial-time determinant implementation was stopped at twelve slots and is retained in the results; it was not a failed identity.

A separate compressed-chord check verifies (10) for all 25 distributions $a + b = 0, 2, 4, 6, 8$ of background Hamiltonian slots. Source, parameter and dependency hashes accompany the saved calculations. The preceding two-point scientific files are kept unchanged. This stage is saved locally without a Git commit, push or archive.

Where this enters the Susskind program. The MSS papers [5, 6] propose a singlet/meson correspondence and call for a quantitative spectrum comparison. The sections read here do not specify a radial reconstruction for the present derivative singlet. Its string-time transfer frequency cannot yet be interpreted as an invariant meson mass. Likewise, the mixed-q representation is a representation of these averaged observables, not a derivation of the entire bulk observable algebra. The present calculation supplies the microscopic correlators needed for that comparison.

The next concrete calculation is the crossed four-point function with the restored q^4 weight, using the existing BINT kernel rather than rederiving its R matrix. Alongside it, we will extract the first small-λ correction to (11) and compare the actual Rindler singlet observables, their time convention and spatial smearing. The correction [7] to the stretched-horizon location remains in force. Finite curvature, finite-N corrections and the on-shell meson coupling require further input.

References

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